

22 Phase change for Qr (content warning: #explosion)

We saw when we studied the Monsanto explosion that the main safety feature on the reactor was a rupture disk that was set to burst when the pressure in the reactor reached 700 psi.

This [video](#) illustrates how a rupture disk works (it is an animation of a pipe fitted with a rupture disk and it has no sound).

According to Fogler, the rupture disk in the Monsanto reactor was fitted with a relief valve with a 2 inch orifice that vented to the atmosphere. The maximum flowrate through the relief valve (from fluid dynamics, calculated for us by Fogler) was hence 830 kg/min.

If the rupture disk had worked properly, once it broke, the pressure in the reactor would have dropped, causing the water to vaporize. The water vapor would've then exited the reactor through the relief valve.

The maximum rate of energy removal (Q_r) due to this process would be:

$$Q_{r, \max} = \dot{m} (\Delta H^{\text{vap}})$$

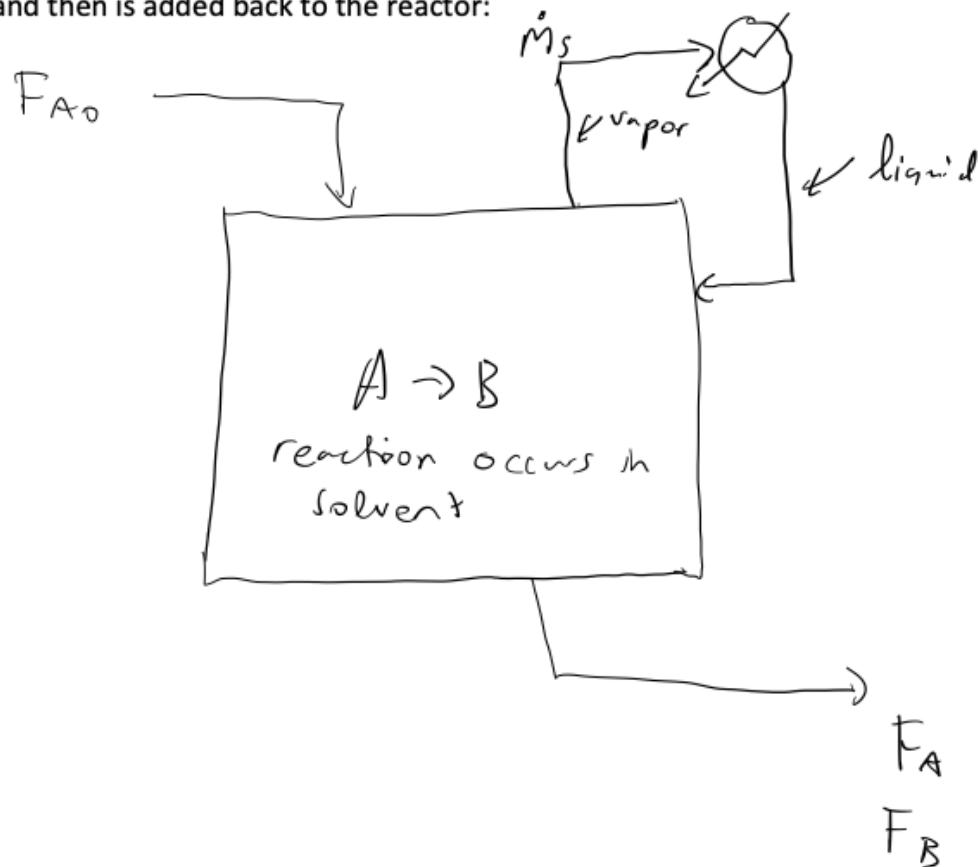
$\uparrow$ max value: $830 \frac{\text{kg}}{\text{min}}$

$\leftarrow 540 \frac{\text{kcal}}{\text{kg}}$

$$Q_{r, \max} = 4.48 \times 10^5 \frac{\text{kcal}}{\text{min}}$$

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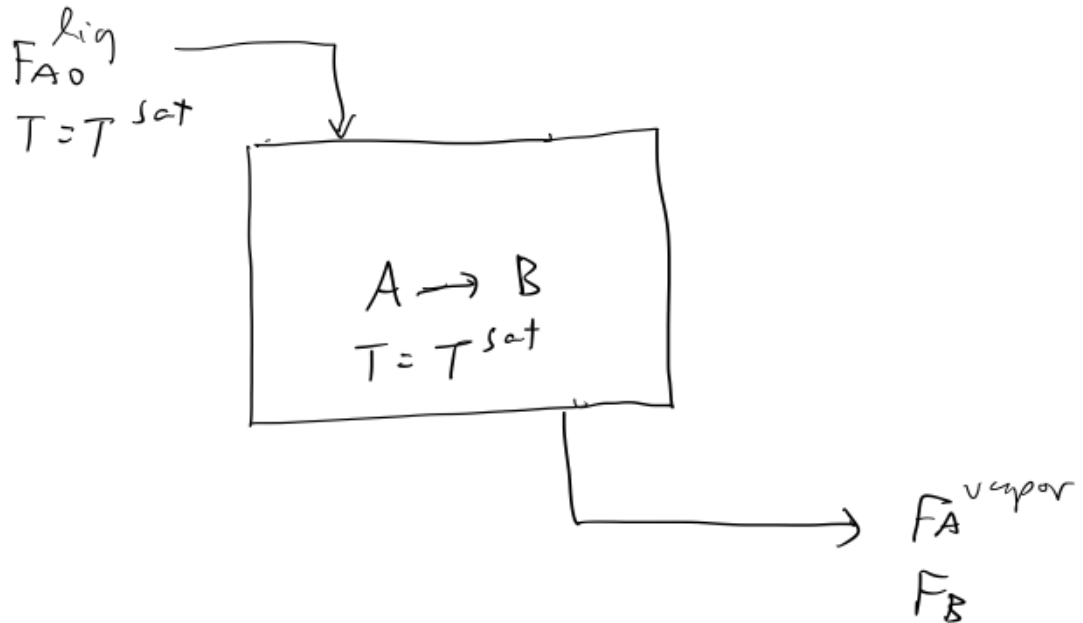
The energy balance for this process is:

$$\begin{aligned}
 0 &= F_{A0} H_{A0} - F_A H_A - F_B H_B - \dot{m}_s H_s^{\text{vapor}} \\
 &+ \dot{m}_s H_s^{\text{liquid}} = F_{A0} H_{A0} - F_A H_A - F_B H_B + \dot{m}_s (-\Delta H_s^{\text{vap}}) \\
 &\quad \underbrace{F_{A0} C_{pA} (T_0 - T)}_{\text{Heat removal by solvent}} \quad \Delta H_s^{\text{vap}} = H_s^{\text{v}} - H_s^{\text{l}} \\
 0 &= F_{A0} C_{pA} T_0 - F_A C_{pA} T + F_{A0} X (-\Delta H_{\text{rxn}}) \\
 &\quad + \dot{m}_s (-\Delta H_s^{\text{vap}}) \\
 0 &= F_{A0} C_{pA} (T_0 - T) + F_{A0} X (-\Delta H_{\text{rxn}}) + \dot{m}_s (-\Delta H_s^{\text{vap}})
 \end{aligned}$$

Where the heat of vaporization provides an additional heat removal term.

Alternatively, one might design a reactor to boil off a product species as a means of distillation. This is called reactive distillation; you can find a short and informative video of reactive distillation [here](#).

Another way to incorporate phase change is to introduce a *liquid* feed to the reactor, which then vaporizes in the reactor due to absorbing the heat generated by the reaction. For example:



The energy balance for this example is:

$$0 = F_{A0}^l H_{A0} - F_A^v H_A - F_B H_B$$

$$0 = F_{A0}^l H_{A0} - (F_{A0}(1-X))^v H_A - F_{A0}X H_B$$

$$0 = F_{A0}^l H_{A0} - F_{A0}^v H_A + F_{A0}^v X H_A - F_{A0}X H_B$$

$$0 = F_{A0}^l H_{A0} - F_{A0}^v H_A + F_{A0}X (-\Delta H_{rxn})$$

$$0 = F_{A0} H_A^{liq} - F_{A0} H_A^{vap} + F_{A0}X (-\Delta H_{rxn})$$

$$0 = F_{A0} (-\Delta H_{A0}^{vap}) + F_{A0}X (-\Delta H_{rxn})$$

Where here, the main difference between the energy balance that we've been working with is that instead of accounting for sensible heat changes between the feed and reactor, we are now accounting for latent heat changes.

this example's $0 = F_{A0} (-\Delta H_A^{\text{vap}}) + F_{A0} X (-\Delta H_{rxn})$

examples we've previously seen's

$$0 = F_{A0} C_{pA} (T_0 - T) + F_{A0} X (-\Delta H_{rxn})$$